

its own hardware capabilities get enhanced locally, it should be able to move back to its own private cloud. That is where the interest in hybrid clouds comes into play.

If a company wants to use a public cloud for some reason, or wants to host some of its workload on it, it would also want to be able to exercise control over it. What we are also seeing is that IT managers are moving the less critical applications on to the public cloud, whereas they manage their more critical applications internally.

Q From tablets and smartphones, users are now moving on to wearable devices. How will wearables impact the cloud computing world?

Wearables are designed to help us do our day-to-day tasks even more easily. They are still in the early stages of adoption, but imagine if you are able to do all your work on a wearable device without depending on any other device! It will change the entire landscape. It will impact the world of cloud computing tremendously. For example, let's say there is a healthcare application that is hosted on the cloud. If you use that application on your wearable device, your doctor can analyse the state of your health in real time. So, if you as a patient come to me, a doctor, I know your entire health history. For instance, currently, patients have no data to present regarding the kind of workouts they do, their effect, the calories burnt, and so on. Sophisticated apps in wearable devices can help to give all such details to a doctor, helping him to analyse his patient's state of health.

Q What exactly is a software defined data centre (SDDC) and how is it changing cloud computing?

As I said, without virtualisation, there is no cloud. A software defined data centre is our architecture for the cloud. We believe that, in the cloud, you need to have a technology that will virtualise network and storage management. You need to have holistic management. These pillars are called 'software defined data centres'. Our goal is to literally virtualise all the services available in the data centre and control them at a software level.

Just as virtualisation is a component of the cloud, a software defined data centre is an architecture for the cloud. It would be difficult to do cloud computing without software defined data centres.

Q It is said that SDDCs are synonymous with private clouds. Do you agree?

I don't agree with this because it is an architecture and it is not restricted to the private cloud. It caters to public clouds as well. It's a technology that you require to build a cloud with all the necessary components, be it management, self-service portals, multi-tenancy or security.

Q Are Indian enterprises ready for SDDCs?

Absolutely. As per an IDC estimate, Indian companies will save around US\$ 3.8 billion by 2020 just with virtualisation.

Why not virtualise the entire data centre and save further? Life is all about speed, today. Everyone wants work to happen at a faster speed and if you can use software to manage your data centre requirements, there's nothing like it!

Q How is the end user computing space getting impacted by SDDCs?

At the end of the day, if the Software-Defined Data Center helps you run the applications because of a massive scale of end user requirements growing significantly by the day, having a model like this will not only boost up the end user computing technology better but will also assist organisations seamlessly deploy the applications with ease and deliver it to the end user anytime, anywhere.

Q What are the steps involved in migrating to SDDCs from the traditional data centres?

At VMware, we say there are three 'journeys' in the process. The first journey is IT production, which looks at virtualising the data centre and cutting down costs. It implies that customers do not want 1000 servers, but just 10-100 servers, especially since servers already have increased compute power these days. Once the capex saving is done, the second journey is about moving into business production, which is all about virtualising a business critical application, like SAP, Hana, et al. The third journey involves moving into complete automation, that is, IT-as-a-service. That is the time you see the entire SDDC stack getting deployed, helping you to virtualise all the components that enable the data centres to provide you IT-as-a-service capabilities. These are the phases that result in the complete adoption of SDDCs.

Q What is your strategy to make SDDCs popular in India amongst IT decision-makers?

We have a track record of IT decision-makers claiming that they could save money by deploying VMware SDDCs. If an IT manager is able to virtualise things beyond just servers, like the network (which is the biggest problem for all the organisations and involves high costs)—if he is able to build those functions into a software and control it, he can save a significant amount of money. That is the need of the hour. The modern-day IT managers want to do more with less.

Q What trends will drive the adoption of SDDCs in the Indian enterprises?

Trends like cloud economics, having to decrease the cost per work load, the willingness to run more optimised data centres compared to anybody else, and the easy availability of applications will define the adoption of SDDCs. The beauty of it all is that developers don't need to develop the applications for SDDCs any differently. They can make vendor-agnostic applications for an operating system, because we have already virtualised the operating system. **END** 